

Károly Pálffy

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[Academic homepage](#)

[LinkedIn](#)

[GitHub](#)

Research-oriented environmental engineer, skilled in phytoplankton ecology, statistics and scientific programming. Developed a process-based phytoplankton community model and an R package for flow cytometric data analysis. Seeking a challenging position in academia to contribute to solutions linking community structure and ecosystem functioning.

Work experience

- | | |
|---------------------------|--|
| Aug 2025
- present | <i>research fellow</i>
HUN-REN Balaton Limnological Research Institute, Tihany, Hungary <ul style="list-style-type: none">● analyze model simulations to link phytoplankton dynamics, nutrient uptake rates and compositional shifts in warming scenarios● expand phytoplankton model to tritrophic food web to assist mesocosm studies |
| April 2021
- July 2025 | <i>research fellow</i>
Centre for Ecological Research (CER), Institute of Aquatic Ecology, Budapest, Hungary <ul style="list-style-type: none">● developed numerical model to simulate plankton dynamics in R● executed flow cytometric analysis of laboratory/field phytoplankton samples, developed R package for flow cytometric data processing● analyzed multivariate data sets (eDNA, field and experimental studies)● managed Danube monitoring in 2023 |
| Aug 2010
- March 2021 | <i>assistant researcher (2010-2011) / research fellow</i>
CER, Balaton Limnological Institute, Tihany, Hungary <ul style="list-style-type: none">● analyzed long-term field and short-term experimental data using R● conducted flow cytometric and microscopic analysis of phytoplankton samples● measured phytoplankton primary production |
| July 2009
- May 2010 | <i>project manager</i>
VTK Innosystem Ltd., Győr, Hungary |
| Jan 2007
- June 2009 | <i>product manager</i>
GyőrSzol Ltd., Győr, Hungary |
| Sep 2004
- Aug 2006 | <i>department engineer</i> – Széchenyi István University, Faculty of Agricultural and Food Sciences, Mosonmagyaróvár, Hungary |

Education

- | | |
|-------------------------|---|
| Sep 2001
- Aug 2004 | <i>PhD</i> – Széchenyi István University, Faculty of Agricultural and Food Sciences (current name: Albert Kázmér Faculty)
Thesis title: Effect of UV radiation on the growth, pigment and hormone content of microalgae (defended in 2011) |
| Sep 1996
- June 2001 | <i>Environmental Engineer MSc</i>
University of Sopron, Sopron, Hungary |

Relevant certificates

- | | |
|----------|---|
| Jan 2026 | <i>IBM Data Science Professional Certificate</i> - Coursera (link embedded) |
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Skills

- statistics
- data science
- numerical modelling
- lab analysis
- field sampling
- driving license

IT skills

- R (RStudio)
- Python (JupyterLab)
- Git (GitHub)

Languages

- English - advanced
- German - intermediate

Teaching experience

- Plankton Ecology (2021-2025, held in English, Eötvös Loránd University, Budapest, Hungary)
- Algal Taxonomy (2016-2018, University of Pécs, Pécs, Hungary)
- Plant physiology (2001-2006, Széchenyi István University, Mosonmagyaróvár, Hungary)

Publication list

ORCID ID: [0000-0003-2015-7972](https://orcid.org/0000-0003-2015-7972) (link embedded)
[Google Scholar](#) (link embedded)